

# TinyDSCNN-48 : Model Architecture & Data Flow

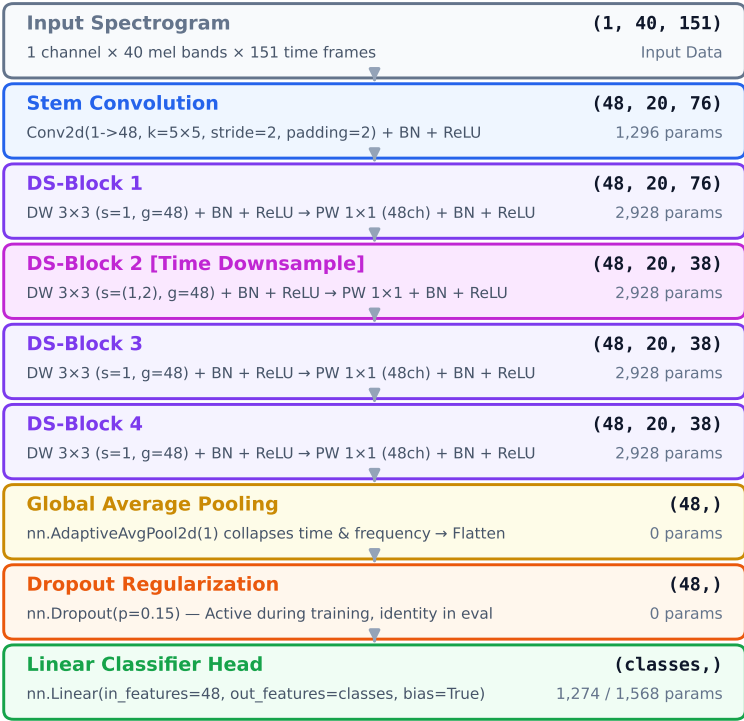
AI 231 (Deep Learning) Machine Exercise 2 — From-Scratch Tiny Voice Command Model

|                                                              |                                                        |                                                     |                                                       |
|--------------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------------|-------------------------------------------------------|
| <b>14,282 PARAMS</b><br>26-class baseline (~14.5k for 32-cl) | <b>44.6 KB ONNX</b><br>Static INT8 quantized footprint | <b>1.5 s WINDOW</b><br>16 kHz Mono (24,000 samples) | <b>0.50 ms CPU</b><br>Single-thread x86 (< 10 ms RPi) |
|--------------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------------|-------------------------------------------------------|

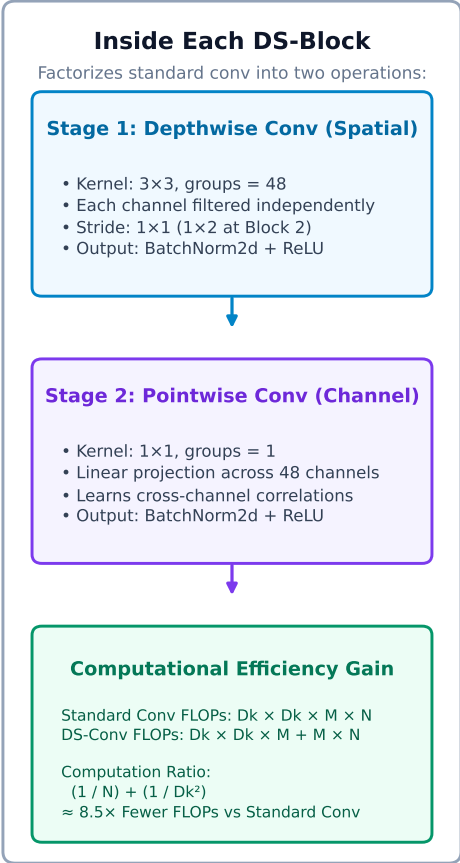
## 1. End-to-End Processing Pipeline



## 2. Layer Flow



## 3. DS-Block Detail



Designed for AI 231 ME2 | From-scratch initialization (Zero pretrained weights) | Exportable to static INT8 ONNX

# TinyDSCNN-48 : Technical Specifications & Provenance

Layer Specifications, Computational Complexity, Edge Footprint, and Academic References

## 1. Layer-by-Layer Dimensions & Parameter Counts

| Layer / Stage      | Layer Operation / Class           | Kernel / Stride / Config          | Output Shape | Params |
|--------------------|-----------------------------------|-----------------------------------|--------------|--------|
| Input Audio        | Log-Mel Spectrogram               | 16 kHz, 25ms win, 10ms hop        | (1, 40, 151) | 0      |
| Stem Conv          | nn.Conv2d + BN + ReLU             | 5×5, s=(2,2), p=2, bias=F         | (48, 20, 76) | 1,296  |
| DS-Block 1 (DW)    | nn.Conv2d (groups=48) + BN + ReLU | 3×3, s=(1,1), p=1, bias=F         | (48, 20, 76) | 528    |
| DS-Block 1 (PW)    | nn.Conv2d + BN + ReLU             | 1×1, s=(1,1), p=0, bias=F         | (48, 20, 76) | 2,400  |
| DS-Block 2 (DW)    | nn.Conv2d (groups=48) + BN + ReLU | 3×3, s=(1,2), p=1, bias=F [s_t=2] | (48, 20, 38) | 528    |
| DS-Block 2 (PW)    | nn.Conv2d + BN + ReLU             | 1×1, s=(1,1), p=0, bias=F         | (48, 20, 38) | 2,400  |
| DS-Block 3 (DW)    | nn.Conv2d (groups=48) + BN + ReLU | 3×3, s=(1,1), p=1, bias=F         | (48, 20, 38) | 528    |
| DS-Block 3 (PW)    | nn.Conv2d + BN + ReLU             | 1×1, s=(1,1), p=0, bias=F         | (48, 20, 38) | 2,400  |
| DS-Block 4 (DW)    | nn.Conv2d (groups=48) + BN + ReLU | 3×3, s=(1,1), p=1, bias=F         | (48, 20, 38) | 528    |
| DS-Block 4 (PW)    | nn.Conv2d + BN + ReLU             | 1×1, s=(1,1), p=0, bias=F         | (48, 20, 38) | 2,400  |
| Global Pooling     | AdaptiveAvgPool2d(1) + Flatten    | Global Average Pooling            | (48, )       | 0      |
| Regularization     | nn.Dropout(p=0.15)                | Active during training            | (48, )       | 0      |
| Classifier Head    | nn.Linear(48 -> 26 classes)       | Dense + Bias                      | (26, )       | 1,274  |
| Total (26 classes) | Baseline Model (Synthetic Run)    | 100 CUDA epochs                   | Logits (26)  | 14,282 |
| Total (32 classes) | Option B Dataset Adaptation       | 32 sub-classes / 19 intents       | Logits (32)  | 14,576 |

## 2. Edge Deployment & Hardware Footprint

|                                                                                                        |                 |                                                                                                              |                   |
|--------------------------------------------------------------------------------------------------------|-----------------|--------------------------------------------------------------------------------------------------------------|-------------------|
| <b>FP32 Uncompressed Model</b><br>PyTorch .pt state_dict / FP32 ONNX weights                           | <b>55.8 KB</b>  | <b>INT8 Quantized ONNX Model</b><br>Nine Conv2D layers and one Linear layer quantized to INT8                | <b>44.6 KB</b>    |
| <b>Runtime RAM (RSS)</b><br>Complete inference process including ONNX Runtime engine                   | <b>73.45 MB</b> | <b>Single-Core CPU Latency</b><br>P50 latency on standard x86 CPU thread; P95 is 0.679 ms                    | <b>0.501 ms</b>   |
| <b>Frontend + Inference Latency</b><br>STFT + Mel filterbank (0.65 ms) + Neural forward pass (0.50 ms) | <b>1.150 ms</b> | <b>Raspberry Pi Execution</b><br>Easily satisfies real-time voice streaming constraint on ARM Cortex-A72/A76 | <b>&lt; 10 ms</b> |

## 3. Academic References & Design Attribution

[1] Zhang, Y., Suda, N., Lai, L., & Chandra, V. (2017). "Hello Edge: Keyword Spotting on Microcontrollers." arXiv:1711.07128. ARM Applied ML Research.  
[2] Howard, A. G., et al. (2017). "MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications." arXiv:1704.04861. Google Inc.  
[3] AI 231 ME2 Adaptation: Scaled to 48 channels, time-strided Block 2, and trained purely from-scratch with zero pretrained weights or foundation models.